

The empirical recurrence for OEIS A278091, recovered from its tail

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Abstract

OEIS A278091 records “Empirical recurrence of order 86” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 144$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 91$. Its last coefficient is $c_{86} = -11390976 \neq 0$, so no further tail satisfies anything shorter; any order-86 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A278091 is “Number of $n \times 5$ 0..1 arrays with every element both equal and not equal to some elements at offset $(-1,0)$ $(-1,1)$ $(0,-1)$ $(0,1)$ or $(1,0)$, with upper left element zero.”. It has offset 1 and begins

0, 27, 351, 6285, 114028, 2066549, 37564017, 682015651, ...

The entry states:

Empirical recurrence of order 86 (see link above)

As of the “Last modified” line on the live entry (Nov 11 2016 20:33:32, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 144$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 144$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 86: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 86.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 288$ exact terms from $n = 5$ on, it returns order 86 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{86} c_i a(n-i),$$

c_1	11	c_{23}	−396459211136307	c_{45}	−625545254068761105	c_{67}	−1810520482088471
c_2	220	c_{24}	−838650277505524	c_{46}	−714440796168483459	c_{68}	−1275102588039746
c_3	−921	c_{25}	1931071982689419	c_{47}	−122153812065054501	c_{69}	859768408380369
c_4	−16873	c_{26}	1973521345386820	c_{48}	870275535049916495	c_{70}	−584297030217378
c_5	37017	c_{27}	−7165794196304421	c_{49}	552282158783920021	c_{71}	874038728931210
c_6	731946	c_{28}	−1265304518220161	c_{50}	−454922027543746812	c_{72}	854361864762
c_7	−938288	c_{29}	20091060607516552	c_{51}	−544346980342625673	c_{73}	153506692991240
c_8	−21189661	c_{30}	−10777099362212185	c_{52}	−251288220628265716	c_{74}	80015351307173
c_9	17099791	c_{31}	−41798686115787839	c_{53}	243254250128655368	c_{75}	−103986073116577
c_{10}	444311937	c_{32}	49707016340617479	c_{54}	585364675157103226	c_{76}	805738582467
c_{11}	−255798904	c_{33}	57100277644890004	c_{55}	51497766875657067	c_{77}	6296509121021
c_{12}	−7080280473	c_{34}	−122986850381111395	c_{56}	−383049725294920019	c_{78}	−2279598241282
c_{13}	3648304574	c_{35}	−17348132671450610	c_{57}	−149098789526017255	c_{79}	−400591238834
c_{14}	88396128549	c_{36}	220633126524755779	c_{58}	73439354524810734	c_{80}	17289154412
c_{15}	−52437344895	c_{37}	−152407740343402121	c_{59}	119495570081929195	c_{81}	−170403512834
c_{16}	−881728938823	c_{38}	−316497246596036282	c_{60}	8167842844181478	c_{82}	4694722316
c_{17}	690135767684	c_{39}	506501468328450834	c_{61}	−49210660348865045	c_{83}	5567662560
c_{18}	7085651411848	c_{40}	371050278947102998	c_{62}	4802349888522464	c_{84}	−4380072960
c_{19}	−7436735340188	c_{41}	−965407317856171906	c_{63}	16190111087193272	c_{85}	240048384
c_{20}	−45562781774809	c_{42}	−381858646976787765	c_{64}	−7311629913085490	c_{86}	−11390976
c_{21}	62131318728689	c_{43}	1115707608091089697	c_{65}	−7020671343827952		
c_{22}	228059089075445	c_{44}	449790833186581312	c_{66}	5557583879215881		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 91$, and it is the recurrence OEIS A278091 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{86} - c_1x^{86-1} - \dots - c_{86}$. Its constant term is $-c_{86} = 11390976$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 86 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 86, so $\deg q = 86$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 86, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 86, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -11390976 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-86 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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